Class 13: Transcriptomics and the analysis of RNA-Seq data

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```

```
Loading required package: generics
```

```
Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo
```

```
Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians
```

```
The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

Background

Today we will perform an RNASeq analysis of the effects of a common sertoid on airway cells.

In particular, dexamethasone (herafter just called "dex") on different airway smooth muscle cell lines (ASM cells).

Data Import:

We need two different inputs:

- **countData**: with genes in rows and experiments in columns
- **colData**: meta data that desscribes the columns in countData

```
counts <- read.csv("airway_scaledcounts.csv", row.names = 1)
metadata <-  read.csv("airway_metadata.csv")
```

Wee peaks at counts and metadata

```
head(counts)
```

```
                SRR1039508 SRR1039509 SRR1039512 SRR1039513 SRR1039516
ENSG00000000003        723        486        904        445       1170
ENSG00000000005          0          0          0          0          0
ENSG00000000419        467        523        616        371        582
ENSG00000000457        347        258        364        237        318
ENSG00000000460         96         81         73         66        118
ENSG00000000938          0          0          1          0          2
                SRR1039517 SRR1039520 SRR1039521
ENSG00000000003       1097        806        604
ENSG00000000005          0          0          0
ENSG00000000419        781        417        509
ENSG00000000457        447        330        324
ENSG00000000460         94        102         74
ENSG00000000938          0          0          0
```

```
heads(metadata)
```

```
   group group_name      value
1      1         id SRR1039508
2      1         id SRR1039509
3      1         id SRR1039512
4      1         id SRR1039513
5      1         id SRR1039516
6      1         id SRR1039517
7      2        dex    control
8      2        dex    treated
9      2        dex    control
10     2        dex    treated
11     2        dex    control
12     2        dex    treated
13     3   celltype     N61311
14     3   celltype     N61311
15     3   celltype    N052611
16     3   celltype    N052611
17     3   celltype    N080611
18     3   celltype    N080611
19     4     geo_id GSM1275862
20     4     geo_id GSM1275863
21     4     geo_id GSM1275866
22     4     geo_id GSM1275867
23     4     geo_id GSM1275870
24     4     geo_id GSM1275871
```

Q1. How many genes are in this dataset? 38694

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many 'control' cell lines do we have? 4

```
table(metadata$dex)
```

```
control treated
      4       4
```

```
sum (metadata$dex == "control")
```

```
[1] 4
```

Differential gene expression

We have 4 replicate drug treated and control (no drug) columns/experiments in our `counts` object.

We want one "mean" value for each gene (rows) in "treated" (drug) and one mean value for each gene in "control" cols.

> Q3. How would you make the above code in either approach more robust? Is there a function that could help here?

Step 1. Find all "control" columns Step 2. Extract these columns to a new object called `control.counts` Step 3. Then calculate the mean value for each gene

Step 1.

```
control.inds <- metadata$dex == "control"
```

Step 2.

```
control.counts <- counts[ , control.inds]
```

Step 3.

```
control.mean <- rowMeans(control.counts)
```

> Q4. Follow the same procedure for the treated samples (i.e. calculate the mean per gene across drug treated samples and assign to a labeled vector called treated.mean)
>
> Q. Now do the same thing for the "treated" columns/experiments...

```
treated.inds <- metadata$dex == "treated"
```

Step 2.

```
treated.counts <- counts[ , treated.inds]
```

Step 3.

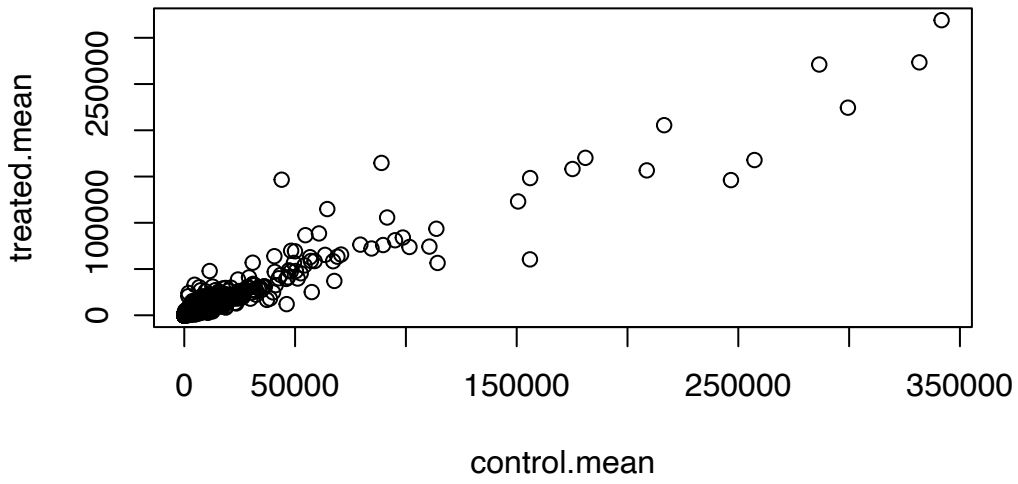
```
treated.mean <- rowMeans(treated.counts)
```

Put these together for easy book-keeping as `meancounts`

```
meancounts <- data.frame(control.mean , treated.mean)
```

Q5. a). Create a scatter plot showing the mean of the treated samples against the mean of the control samples. Your plot should look something like the following. A quick plot

```
plot(meancounts)
```



Q5 (b).You could also use the ggplot2 package to make this figure producing the plot below. What geom_?() function would you use for this plot? point

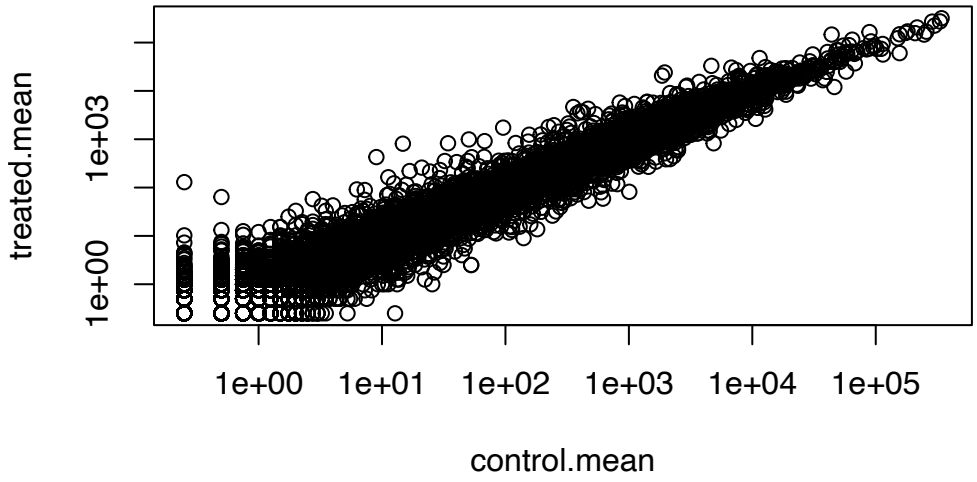
Q6. Try plotting both axes on a log scale. What is the argument to plot() that allows you to do this? plot(meancounts, log="xy")

Let's log transform this count data:

```
plot(meancounts, log="xy")
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted
from logarithmic plot
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted
from logarithmic plot
```



N.B We most often use log2 for this type of data as it makes the interpretation much more straightforward.

Treated/Control is often called "fold-change"

If there was no change we would have a log2-fc of zero

```
log2(10/10)
```

```
[1] 0
```

If we had double the amount of transcript around

```
log2(20/10)
```

```
[1] 1
```

If we had half as much transcript around we would have a log2-fc of -1

```
log2(5/10)
```

```
[1] -1
```

> Q. Calculate a log2 fold change value for all our genes and add it as a new column to our `meancounts` object.

```
meancounts$log2f <- log2(meancounts$treated.mean/meancounts$control.mean)
head(meancounts)
```

```
                control.mean treated.mean       log2f
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000005         0.00         0.00         NaN
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000938         0.75         0.00        -Inf
```

```
zero.vals <- which(meancounts[,1:2]==0, arr.ind=TRUE)

to.rm <- unique(zero.vals[,1])
mycounts <- meancounts[-to.rm,]
head(mycounts)
```

```
                control.mean treated.mean       log2f
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000971      5219.00      6687.50  0.35769358
ENSG00000001036      2327.00      1785.75 -0.38194109
```

> Q7. What is the purpose of the arr.ind argument in the which() function call above? Why would we then take the first column of the output and need to call the unique() function?

```
up.ind <- mycounts$log2fc > 2
down.ind <- mycounts$log2fc < (-2)

sum(up.ind)
```

```
[1] 0
```

```
sum(down.ind)
```

```
[1] 0
```

> Q8. Using the up.ind vector above can you determine how many up regulated genes we have at the greater than 2 fc level? 250 Q9. Using the down.ind vector above can you determine how many down regulated genes we have at the greater than 2 fc level? 367 Q10. Do you trust these results? Why or why not? No, I do not trust these results on its own because a gene could show a large fold chanage for multiple reasons. There is no statistical testing so false positives could occur.

There are some "funky" log2fc values (NaN and -Inf) here that come about whenever we have 0 mean count values. Typically we would remove these genes from any further analysis - as we can't say anything about them if we have no data for them.

PCA

```
dds <- DESeqDataSetFromMatrix(countData = counts,
                        colData = metadata,
                        design = ~dex)
```

```
converting counts to integer mode

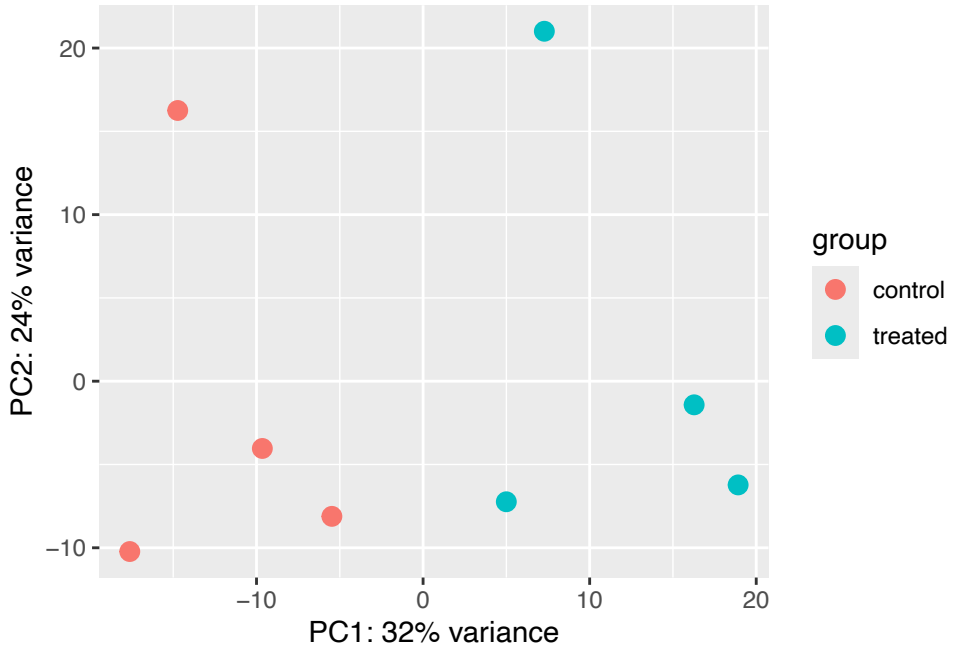
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```
dds
```

```
class: DESeqDataSet
dim: 38694 8
metadata(1): version
assays(1): counts
rownames(38694): ENSG00000000003 ENSG00000000005 ... ENSG00000283120
  ENSG00000283123
rowData names(0):
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(4): id dex celltype geo_id
```

```
vsd <- vst(dds, blind = FALSE)
plotPCA(vsd, intgroup = c("dex"))
```

```
using ntop=500 top features by variance
```




```
pcaData <- plotPCA(vsd, intgroup=c("dex"), returnData=TRUE)
```

```
using ntop=500 top features by variance
```

```
head(pcaData)
```

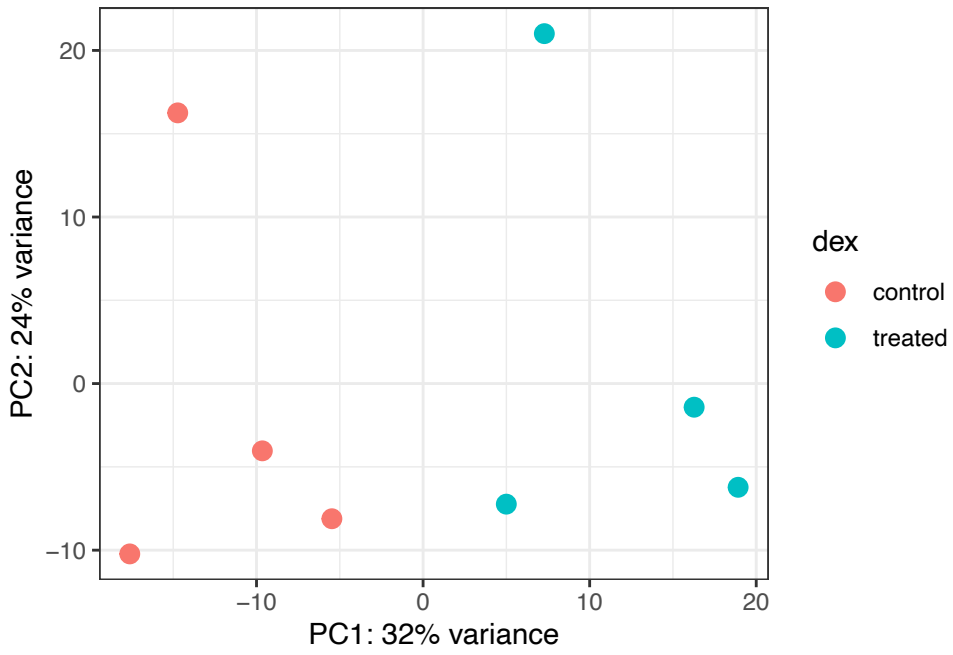
```
                PC1        PC2   group       name         id     dex celltype
SRR1039508 -17.607922 -10.225252 control SRR1039508 SRR1039508 control   N61311
SRR1039509   4.996738  -7.238117 treated SRR1039509 SRR1039509 treated   N61311
SRR1039512  -5.474456  -8.113993 control SRR1039512 SRR1039512 control  N052611
SRR1039513  18.912974  -6.226041 treated SRR1039513 SRR1039513 treated  N052611
SRR1039516 -14.729173  16.252000 control SRR1039516 SRR1039516 control  N080611
SRR1039517   7.279863  21.008034 treated SRR1039517 SRR1039517 treated  N080611
              geo_id sizeFactor
SRR1039508 GSM1275862  1.0193796
```

```
SRR1039509 GSM1275863  0.9005653
SRR1039512 GSM1275866  1.1784239
SRR1039513 GSM1275867  0.6709854
SRR1039516 GSM1275870  1.1731984
SRR1039517 GSM1275871  1.3929361
```

```r
# Calculate percent variance per PC for the plot axis labels
percentVar <- round(100 * attr(pcaData, "percentVar"))
```

```r
library(ggplot2)
```

```r
ggplot(pcaData) +
  aes(x = PC1, y = PC2, color = dex) +
  geom_point(size =3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  coord_fixed() +
  theme_bw()
```



DESeq analysis

Let's do this analysis with an estimate of statistical significance using **DESeq2** package

```
library(DESeq2)
citation("DESeq2")
```

```
To cite package 'DESeq2' in publications use:

  Love, M.I., Huber, W., Anders, S. Moderated estimation of fold change
  and dispersion for RNA-seq data with DESeq2 Genome Biology 15(12):550
  (2014)

A BibTeX entry for LaTeX users is

  @Article{,
    title = {Moderated estimation of fold change and dispersion for RNA-
seq data with DESeq2},
    author = {Michael I. Love and Wolfgang Huber and Simon Anders},
    year = {2014},
    journal = {Genome Biology},
    doi = {10.1186/s13059-014-0550-8},
    volume = {15},
    issue = {12},
    pages = {550},
  }
```

DESeq (like many bioconductor packages) want it's inpput data in a very specific way.

```
dds <- DESeqDataSetFromMatrix(countData = counts,
                        colData = metadata,
                        design = ~dex)
```

```
converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```
dds
```

```
class: DESeqDataSet
dim: 38694 8
metadata(1): version
assays(1): counts
```

```
rownames(38694): ENSG00000000003 ENSG00000000005 ... ENSG00000283120
  ENSG00000283123
rowData names(0):
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(4): id dex celltype geo_id
```

Run the DESeq analysis pipeline

The main function `DESeq()`

```
dds <- DESeq(dds)
```

```
estimating size factors
```

```
estimating dispersions
```

```
gene-wise dispersion estimates
```

```
mean-dispersion relationship
```

```
final dispersion estimates
```

```
fitting model and testing
```

```
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 6 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj
```

```
                <numeric>
ENSG00000000003  0.163017
ENSG00000000005        NA
ENSG00000000419  0.175937
ENSG00000000457  0.961682
ENSG00000000460  0.815805
ENSG00000000938        NA
```

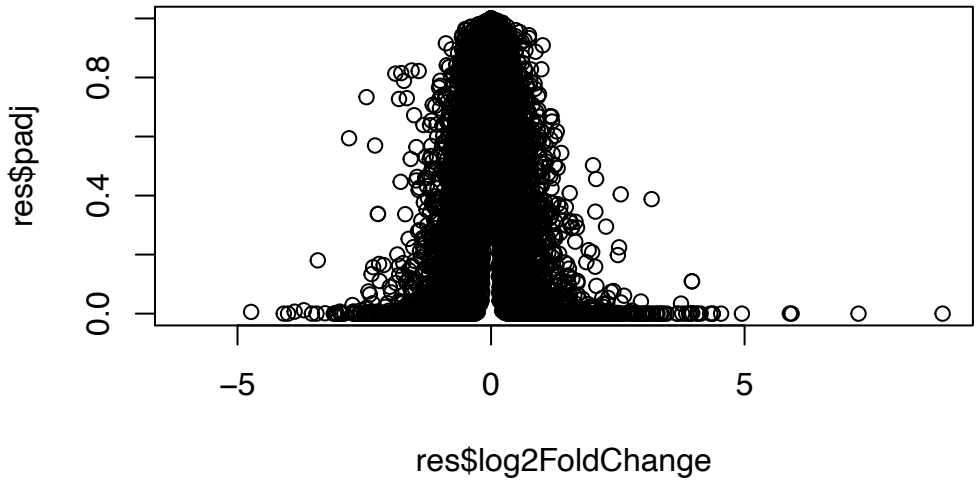
```
36000 * 0.05
```

```
[1] 1800
```

Volcano Plot

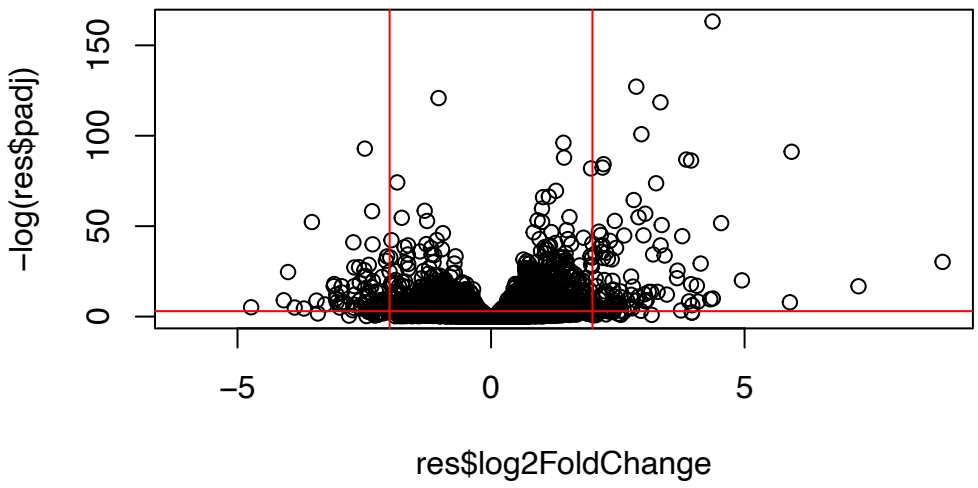
This is a main summary results figure from these kinds of studies. It is a plot of Log2 fold-change vs. (Adjusted) P-value.

```
plot(res$log2FoldChange,
     res$padj)
```



Again this y-axis is highly needs log transforming

```
plot(res$log2FoldChange,
     -log(res$padj))
abline(v=-2, col="red")
abline(v=+2, col="red")
abline(h=-log(0.05), col="red")
```



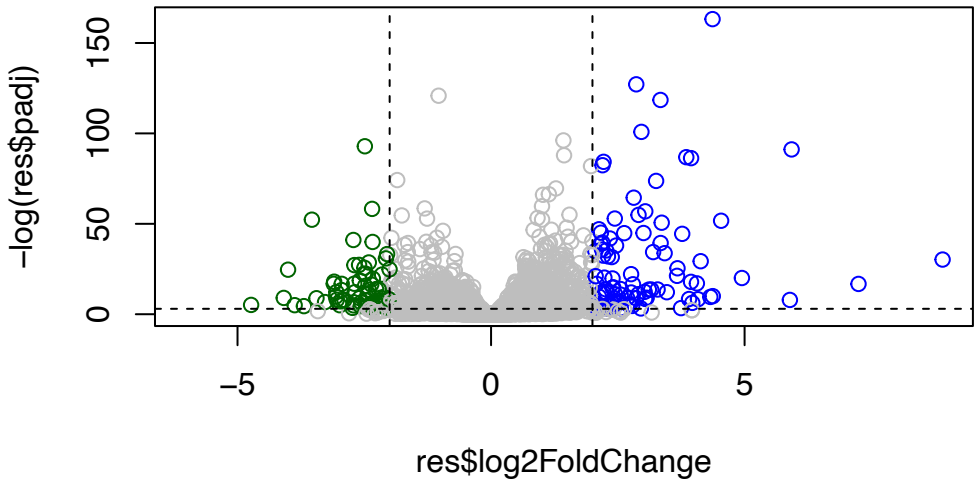
Adding some color annotation

Start with a default base color "grey"

```
# Custom colors
mycols <- rep("grey", nrow(res))
mycols [res$log2FoldChange > 2] <- "blue"
mycols [res$log2FoldChange < -2] <- "darkgreen"
mycols [res$padj >= 0.05] <- "grey"

# Volcano plot
plot(res$log2FoldChange,
     -log(res$padj),
     col=mycols)
```

```
# Cut-off dashed lines
abline(v=c(-2,+2), lty=2)
abline(h=-log(0.05), lty=2)
```

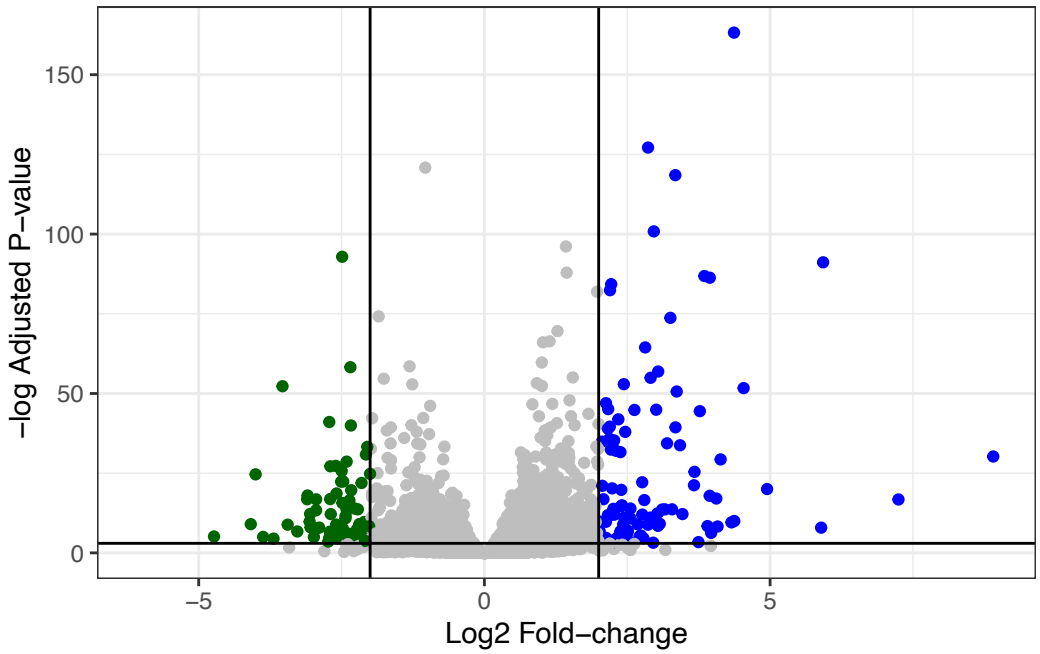


Q. Make a presentation quality ggplot version of this plot. Include cleaer axis labels, a clean theme, your custom colors, cut-off lines and a plot title.

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(colour = mycols) +
  labs(x="Log2 Fold-change",
       y="-log Adjusted P-value") +
geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.05)) +
theme_bw()
```

```
Warning: Removed 23549 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Save our results

Write a CSV file

```
write.csv(res, file="results.csv")
```

Add annotation data

We need to add missing annotation data to our main `res` results object. This includes the common gene "symbol."

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 6 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
```

```
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj
                  <numeric>
ENSG00000000003  0.163017
ENSG00000000005        NA
ENSG00000000419  0.175937
ENSG00000000457  0.961682
ENSG00000000460  0.815805
ENSG00000000938        NA
```

We will use R and bioconductor to do this "ID mapping"

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

Let's see what databases we can use for translation/mapping

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

We can use the `mapIds()` function now to "translate" between any of these databases.

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",   # The format of our genenames
                     column="SYMBOL",     # The new format we want to add
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 7 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj      symbol
                <numeric> <character>
ENSG00000000003  0.163017      TSPAN6
ENSG00000000005        NA        TNMD
ENSG00000000419  0.175937        DPM1
ENSG00000000457  0.961682       SCYL3
ENSG00000000460  0.815805       FIRRM
ENSG00000000938        NA         FGR
```

Q11. Run the mapIds() function two more times to add the Entrez ID and UniProt accession and GENENAME as new columns called res$entrez$, res$uniprot and res$genename.

Q. Also add "ENTREZID", "GENENAME" IDs to our `res` object:

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="ENTREZID",
                     keytype="ENSEMBL",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$uniprot <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="UNIPROT",
                     keytype="ENSEMBL",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$genename <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="GENENAME",
                     keytype="ENSEMBL",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 10 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj      symbol      entrez     uniprot
                <numeric> <character> <character> <character>
ENSG00000000003  0.163017      TSPAN6        7105  A0A087WYV6
ENSG00000000005        NA        TNMD       64102      Q9H2S6
ENSG00000000419  0.175937        DPM1        8813      H0Y368
ENSG00000000457  0.961682       SCYL3       57147      X6RHX1
ENSG00000000460  0.815805       FIRRM       55732      A6NFP1
ENSG00000000938        NA         FGR        2268      B7Z6W7
                           genename
                        <character>
ENSG00000000003          tetraspanin 6
ENSG00000000005            tenomodulin
ENSG00000000419 dolichyl-phosphate m..
ENSG00000000457 SCY1 like pseudokina..
ENSG00000000460 FIGNL1 interacting r..
ENSG00000000938 FGR proto-oncogene, ..
```

Save annotated results to a CSV file

```
write.csv(res, file="results_annotated.csv")
```

Pathway analysis

What known biological pathways do our differently expressed genes overlap with (i.e. play a role in)?

There are lot's of bioconductor packages to do this tyoe of analysis.

We will use one of the oldest called **gage** along with **pathview** to render nice pics of the pathways we find.

We can install these with the command: `BiocManager::install( c("pathview", "gage", "gageData") )`

```
library(pathview)
library(gage)
library(gageData)
```

Have a wee peak what is in `gageData`

```
# Examine the first 2 pathways in this kegg set for humans
data(kegg.sets.hs)
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"
```

The main `gage()` function that does the actual work wants a simple vector as input.

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$symbol
head(foldchanges)
```

```
     TSPAN6        TNMD        DPM1       SCYL3       FIRRM         FGR
-0.35070296          NA  0.20610728  0.02452701 -0.14714263 -1.73228897
```

The KEGG database uses ENTREZ ids so we need to provide these in our input vector for **gage**:

```
names(foldchanges) <- res$entrez
```

Now we can run `gage()`

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

What is in the output object `keggress`

```
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
# Look at the first three down (less) pathways
head(keggres$less, 3)
```

```
                                    p.geomean stat.mean        p.val
hsa05332 Graft-versus-host disease 0.0004250607 -3.473335 0.0004250607
hsa04940 Type I diabetes mellitus  0.0017820379 -3.002350 0.0017820379
hsa05310 Asthma                    0.0020046180 -3.009045 0.0020046180
                                         q.val set.size         exp1
hsa05332 Graft-versus-host disease 0.09053792       40 0.0004250607
hsa04940 Type I diabetes mellitus  0.14232788       42 0.0017820379
hsa05310 Asthma                    0.14232788       29 0.0020046180
```

We can use the **pathview** function to render a figure of any of these pathways along with annotation for our DEGs

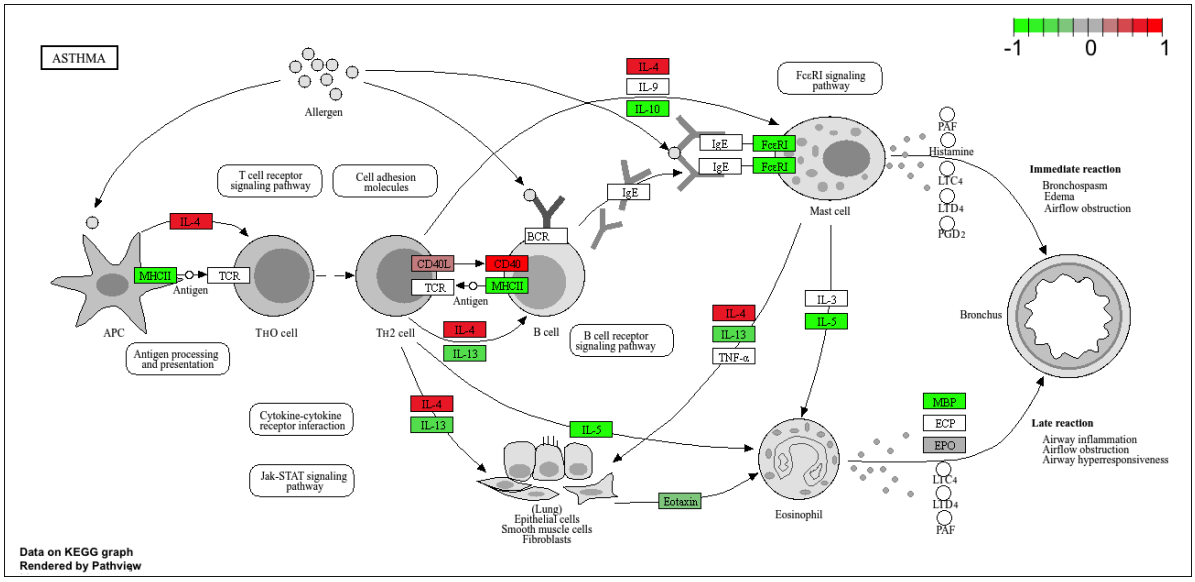
Let's see the hsa05310 Asthma pathway with our DEFs colored up:

```
pathview(gene.data=foldchanges, pathway.id="hsa05310")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/amynguyen/Desktop/BIMM143/class 13
```

```
Info: Writing image file hsa05310.pathview.png
```



> Q12. Can you do the same procedure as above to plot the pathview figures for the top 2 down-reguled pathways?

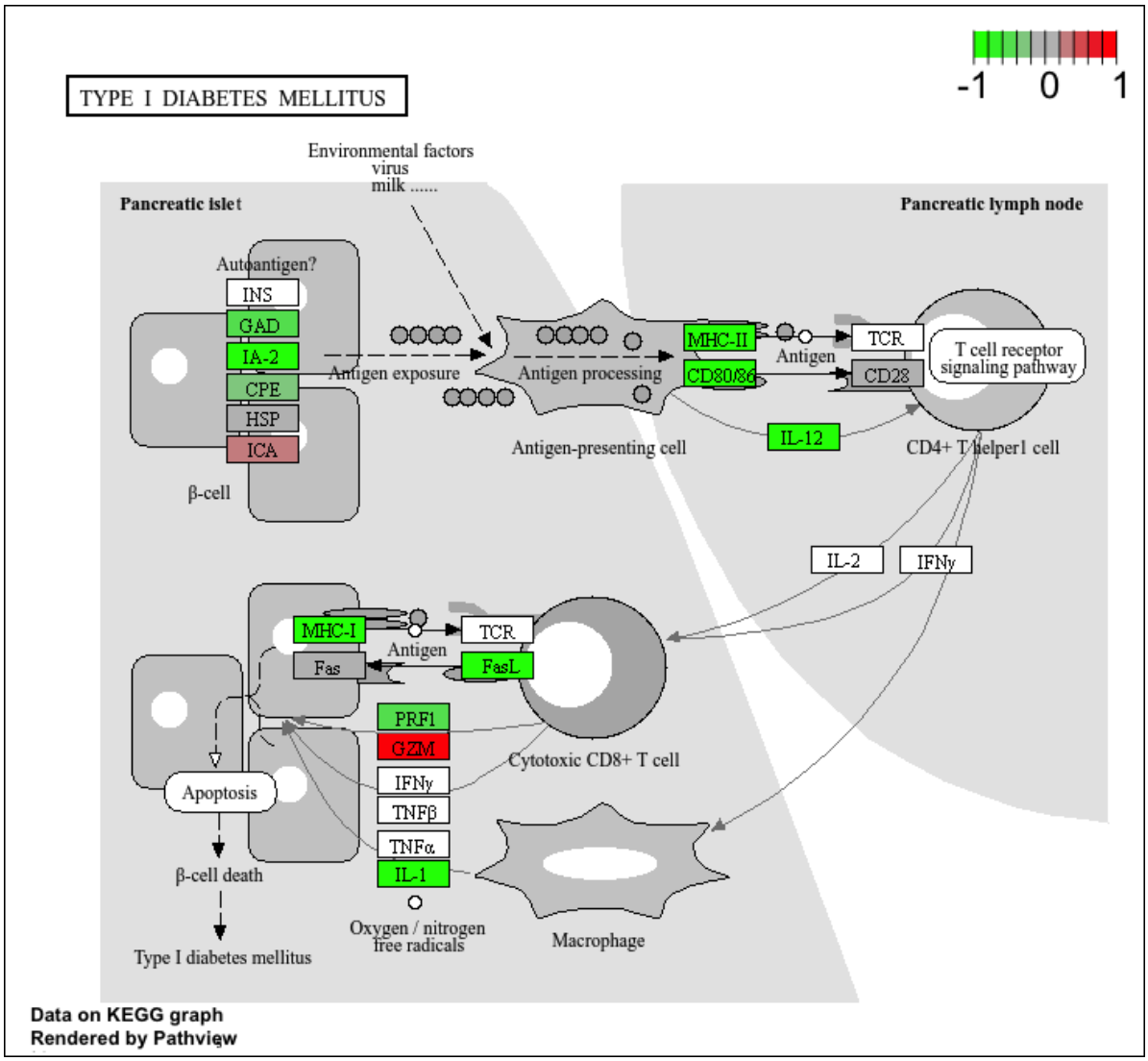
Q. Can you render and insert here the pathway figure for "Graft-versus-host disease" and "Type I diabetes"?

```
pathview(gene.data=foldchanges, pathway.id="hsa04940")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/amynguyen/Desktop/BIMM143/class 13
```

```
Info: Writing image file hsa04940.pathview.png
```

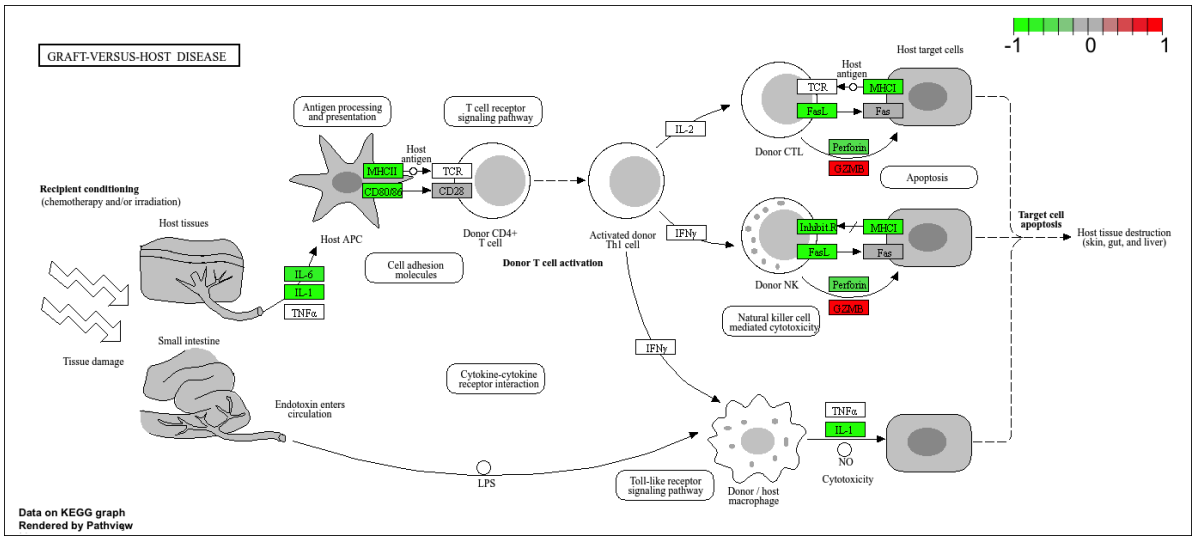



```
pathview(gene.data=foldchanges, pathway.id="hsa05332")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/amynguyen/Desktop/BIMM143/class 13
```

```
Info: Writing image file hsa05332.pathview.png
```



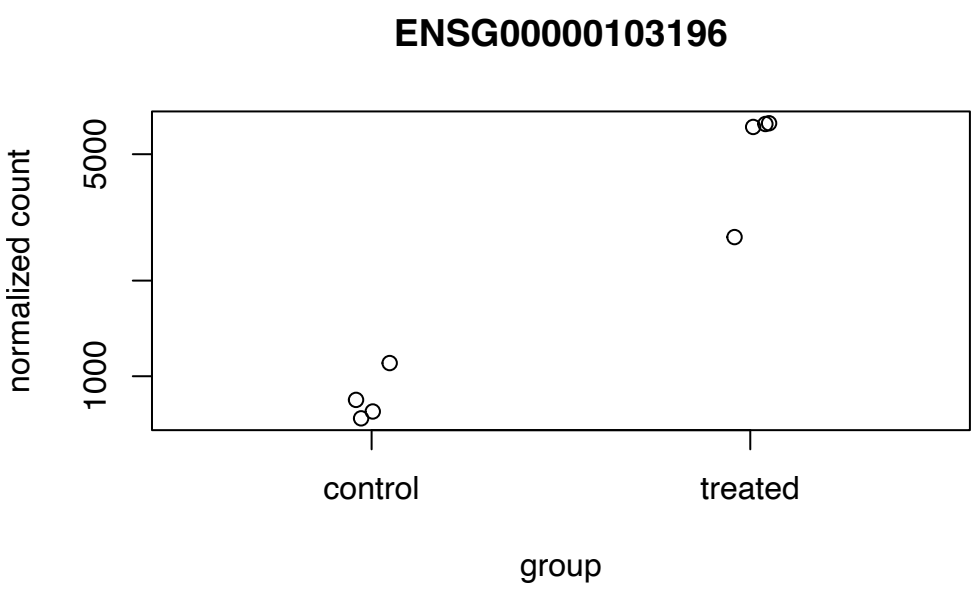

```
i <- grep("CRISPLD2", res$symbol)
res[i,]
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 1 row and 10 columns
                 baseMean log2FoldChange     lfcSE      stat      pvalue
                <numeric>      <numeric> <numeric> <numeric>   <numeric>
ENSG00000103196   3096.16        2.62603  0.267454   9.81865 9.35839e-23
                      padj      symbol      entrez     uniprot
                 <numeric> <character> <character> <character>
ENSG00000103196 3.37459e-20    CRISPLD2       83716  A0A140VK80
                           genename
                        <character>
ENSG00000103196 cysteine rich secret..
```

```
rownames(res[i,])
```

```
[1] "ENSG00000103196"
```

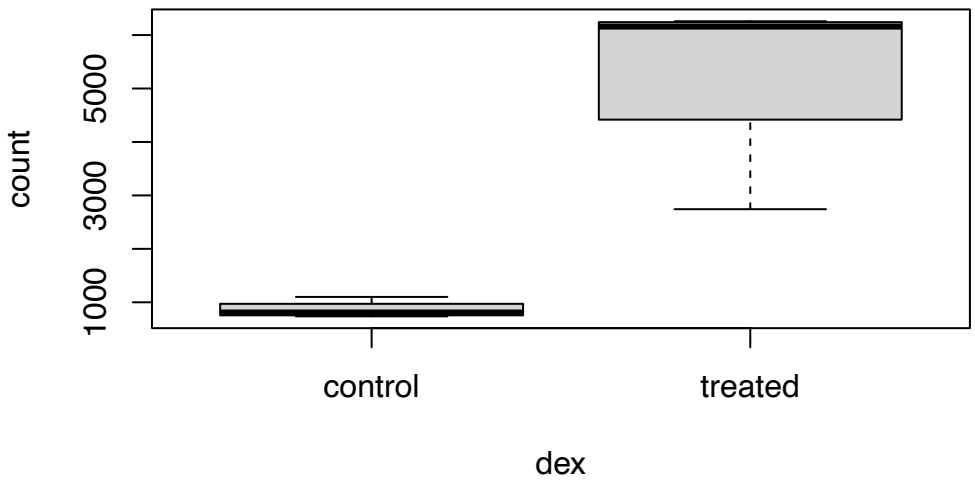
```
plotCounts(dds, gene="ENSG00000103196", intgroup="dex")
```



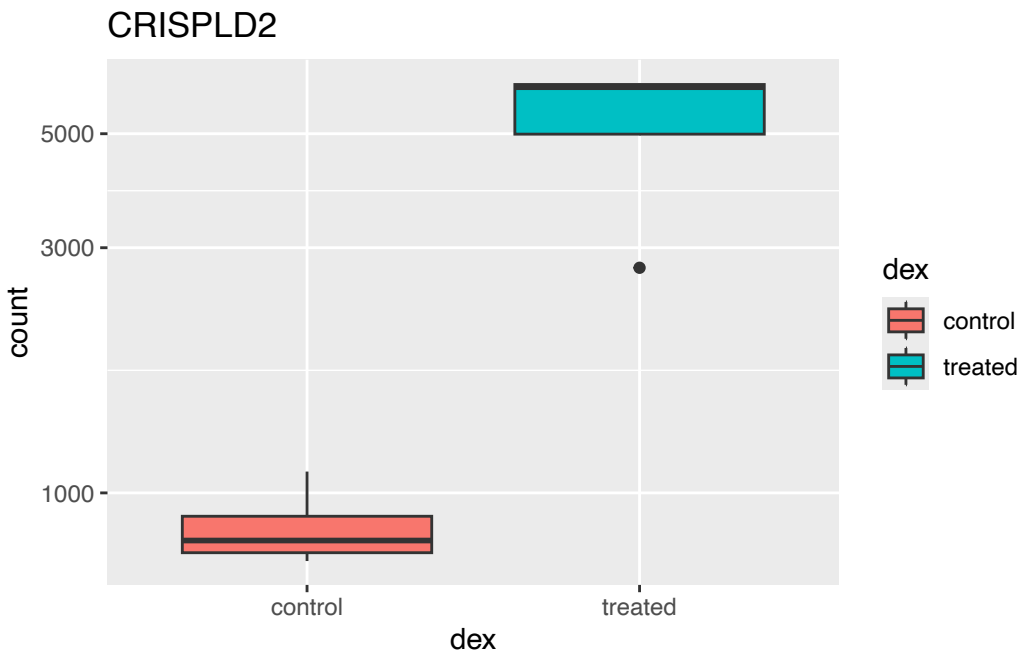

```
# Return the data
d <- plotCounts(dds, gene="ENSG00000103196", intgroup="dex", returnData=TRUE)
head(d)
```

```
               count     dex
SRR1039508  774.5002 control
SRR1039509 6258.7915 treated
SRR1039512 1100.2741 control
SRR1039513 6093.0324 treated
SRR1039516  736.9483 control
SRR1039517 2742.1908 treated
```

```
boxplot(count ~ dex , data=d)
```




```
library(ggplot2)
ggplot(d, aes(dex, count, fill=dex)) +
  geom_boxplot() +
  scale_y_log10() +
  ggtitle("CRISPLD2")
```


CRISPLD2
5000
3000
1000
count
control
treated
dex
dex
control
treated